Short Report on DON Concentration Prediction Using Hyperspectral Data

1. Preprocessing Steps and Rationale

This project aimed to predict **Deoxynivalenol (DON)** levels in corn samples using hyperspectral imaging data. The dataset consisted of 500 samples with 448 spectral features and a target variable vomitoxin_ppb.

Key Preprocessing Steps:

- **Data Cleaning**:
  - The hsi_id column (sample ID) was dropped as it held no predictive value.
  - Missing values were imputed using column-wise mean, ensuring data consistency.
- **Feature Scaling**:
  - All spectral features were normalized to a [0, 1] range using MinMaxScaler.
  - Scaling was essential to improve convergence and ensure all features contributed equally during model training.
- **Feature Reduction**:
  - Although PCA was not explicitly used, model complexity was controlled by selecting relevant features and avoiding redundancy through correlation checks.

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2. Model Selection, Training, and Evaluation

Model Architecture:

- A **Feedforward Neural Network (DNN)** was implemented with:
  - Input Layer: 448 nodes (one per spectral feature).
  - Hidden Layers: Two hidden layers with 128 and 64 ReLU-activated units.
  - Output Layer: Single node for DON prediction (regression task).

Training Strategy:

- The model was trained for **50 epochs** using the **Adam optimizer** and **Mean Squared Error (MSE)** as the loss function.
- Training/Validation split: 80/20
- Batch size: 32

Evaluation Metrics:

Metric	Value
Mean Absolute Error (MAE)	**4018.47**
Root Mean Squared Error (RMSE)	**13164.73**
R^2 Score	**-0.01**

The low R^2 and high error values suggest that the model underfits and further refinement is required.

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3. Insights from Model Interpretability (LIME)

To explain the model's predictions and uncover the influence of individual features, **LIME (Local Interpretable Model-agnostic Explanations)** was applied.

- A single sample was used to generate a local surrogate explanation.
- LIME successfully identified top features contributing to the DON prediction.
- The explanation was saved to outputs/lime_explanation_sample0.html and confirmed feature importance rankings.

This insight is crucial for trust and transparency in model predictions.

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4. Key Findings and Recommendations

Findings:

- The neural network successfully learned to map hyperspectral features to DON levels but demonstrated limited generalization (as reflected in MAE/R^2).
- The pipeline successfully produced predictions and offered explanations via LIME.
- Output visualizations like the **actual vs. predicted scatter plot** confirmed prediction dispersion.

Recommendations:

- **Improve Model Generalization**:
    - Experiment with **hyperparameter tuning** (e.g., learning rate, architecture depth).
    - Test **regularization techniques** like dropout or L2 norm.
- **Explore Alternative Models**:
    - Try XGBoost, Random Forest, or Ensemble methods for regression.
- **Dimensionality Reduction**:
    - Apply PCA or autoencoders to further reduce feature noise and improve learning.
- **Dataset Expansion**:
    - More labeled hyperspectral samples could help the model generalize better.

Conclusion

This project successfully built a complete end-to-end DON prediction pipeline using hyperspectral data. It incorporated preprocessing, model training, performance evaluation, and interpretability with LIME. Although the current model needs further tuning for better accuracy, the overall framework is modular, explainable, and ready for deployment or scaling.